Assignment -1

Solve all the problems.

1. There are two bags A and B. A contains n white balls and 2 black balls. B contains 2 white and n black balls. One of the two bags is selected at random and two balls are drawn from it without replacement. If both the balls drawn are white and the probability that the bag A was used to draw the balls is 6/7. Find the value of n.

2. A bag contains 17 counters marked with the numbers 1 to 17. A counter is drawn and replaced; a second drawing is then made. What is the probability that:
(i) the first number drawn is even & the second odd?
(ii) The first number is odd and the second even?

3. If $p(x) = \begin{cases} \frac{x}{15} : x = 1,2,3,4,5 \\ 0 : \text{elsewhere} \end{cases}$

Find (i) $P(X=1 \text{ or } 2)$ (ii) $P\{\frac{1}{2} < x < \frac{5}{2} / x > 1\}$

4. A continuous random variable X has a p.d.f $f(x) = 3x^2$, $0 \leq x \leq 1$.

Find a and b such that

(i) $P(x \leq a) = P(x > a)$ (ii) $P(x > b) = 0.05$